

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in validated interval arithmetic

Ethan Knox

Independent Researcher, Champaign, IL
ethank5149@gmail.com

Preliminary PhD Research Proposal

October 6, 2026

Abstract

Computing the reachable set of a six-degree-of-freedom hypersonic entry vehicle is intractable for grid-based dynamic programming, and a bound produced by a numerical solver inherits the solver's accuracy. We propose an analytical framework for certified outer bounds on such reachable sets. The dynamics are embedded in a space of occupation measures, where they become a linear constraint, and the transcendental aerodynamics are enclosed in intervals, so that the vector field is a differential inclusion evaluated in validated interval arithmetic. The new content is the thin-atmosphere limit. In this limit the occupation measures converge, but their limit forgets every atmospheric pass: a pass lasts a vanishing time, and yet it is where the trajectories of the family are told apart. We read the loss on the energy dissipation along the rescaled time axis, where the concentration-compactness principle of Lions applies, and extract a profile decomposition: a low-dimensional skeleton, formed by the arcs above the atmosphere and the equilibrium glide; one boundary-layer bubble per atmospheric pass; and a residual. The target statement is that the reachable set is contained in that of the limit system formed by the skeleton and the bubbles, thickened by a ball whose radius is explicit. The constants that size the bound are to be derived in closed form from the physical corridor and the aerodynamic interval widths. The bound then holds uniformly over the entire admissible family, in the minimal physical state space, and it is re-evaluated rather than re-solved when the vehicle or the corridor changes.

1 The analytical problem

The object to be bounded is the reachable set of an atmospheric entry vehicle, that is, every state it can be driven to under admissible control. This is what a safety analysis needs enclosed. We defer the flight-mechanics specifics to [Section 3](#), which explains why entry is the right instance and describes its state space. What matters first is that the problem is an analytical one before it is an aerospace one, and that the existing methods fall short of it for two independent reasons: the dimension of the state, discussed with the application in [Section 3](#), and the provenance of the guarantee, discussed in [Section 1.2](#). There are therefore two problems to solve, not one.

The approach has four steps. The first is the *embedding*: the coupled flight dynamics are written with the aerodynamics enclosed in intervals, so that they form a differential inclusion. The second is the *relaxation*: the inclusion is lifted to a linear problem on occupation measures ([Section 1.1](#)). The third is the *decomposition*: in the thin-atmosphere limit the family of measures loses exactly the events that matter, and we recover them through a profile decomposition into a skeleton, bubbles and a residual ([Section 2](#)). The fourth is the *certification*: the constants of the resulting bound are derived in closed form and evaluated in validated interval arithmetic ([Sections 1.2](#) and [2.4](#)). Six objectives carry the program. They are stated where each arises and gathered in [Section 4](#). One estimate is elementary enough to be carried out here, and we do so in [Section 2.5](#).

1.1 Reachable sets as occupation measures

We embed the flight dynamics in a space of occupation measures. Consider the control system $\dot{x} = f(x, c, t)$, with the state x in the flight corridor $\mathcal{X}$ and the control c in the admissible set $\mathcal{U}$. Along a trajectory, the time spent in each set defines an occupation measure μ on $[0, T] \times \mathcal{X} \times \mathcal{U}$. In terms of μ the nonlinear flight mechanics become *linear*: the Liouville equation is a linear constraint relating μ to an initial measure and to a terminal measure μ_T , and the reachable set at time T is the union of the supports $\text{supp } \mu_T$ over all measures that satisfy it. An outer bound is obtained from the dual problem. A pair of functions (Φ, Ψ) , the first non-decreasing along every admissible trajectory and at least one on the initial set, the second majorizing the first at time T , encloses the reachable set in the superlevel set $\{\Psi \geq 1\}$. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [\[1, 2\]](#). It was developed for regions of attraction and invariant sets [\[3, 4\]](#) and, for hybrid systems, for reachable sets and optimal control, where a measure on each guard carries the mass from one mode to the next [\[5, 6\]](#).

The embedding buys linearity at the price of a relaxation. A measure that satisfies the Liouville constraint is generated by a family of trajectories of the *convexified* inclusion $\dot{x} \in \text{conv } f(x, \mathcal{U}, t)$, and not necessarily by a single trajectory of the original system [\[3, Lemma 3\]](#). Since the relaxed reachable set contains the true one, an outer bound on the first is an outer bound on the second, which is the direction we need.

The same relaxation admits the interval envelope. The true aerodynamics are transcendental and known only to within an envelope, so we enclose them in intervals. A vector field selected from an interval is a set-valued map, and the resulting differential inclusion, not the nominal field, is the object that the measure must describe [\[7, 8\]](#). If the selection is written as an additional box-valued input, the inclusion is a control system with a bounded disturbance, and at fixed ε its Liouville constraint is the standard one. What is not standard is to carry the outer approximation through the singular limit of [Section 2](#), uniformly over the selections.

The nearest prior art is the occupation-measure theory of singularly perturbed inclusions [\[9\]](#). Separating time scales is a standard device in flight mechanics [\[10\]](#), and the classical reduction, which sets $\varepsilon = 0$, requires the fast motion to settle. The theory of [\[9\]](#) does not. It defines an invariant

measure for the fast motion without assuming uniqueness of solutions, and it lets the slow dynamics be driven by these invariant measures. Its hypotheses show where the entry problem departs from it. The linear growth condition on the field only excludes finite-time blow-up, and it is automatic on a bounded corridor. The singular-perturbation theorem, however, assumes a uniform bound on the fast state along the whole ε -family [9, 11]. The natural fast state of entry is the altitude measured in scale heights. On a corridor of fixed altitude extent this variable ranges over an interval of length proportional to $1/\varepsilon$, the aerodynamic terms grow exponentially in it, and a skip that rises through a fixed altitude carries it out of every bounded set. The hypothesis therefore holds for motion that stays within a bounded number of scale heights of the glide, and it fails exactly where an atmospheric pass concentrates (Section 2.1).

Objective 1. Liouville equation for the interval inclusion

At fixed ε the interval inclusion $\dot{x} \in \mathcal{F}_\varepsilon(x, c, t)$ is a control system with a box-valued disturbance, and its Liouville constraint is standard. So are the bounds that hold uniformly in ε on mass, on support and, through the energy balance, on dissipation. Two things are not standard. First, we must construct a certified upper and lower envelope of the atmospheric density, which fixes $\mathcal{F}_\varepsilon$ concretely. Second, we must identify the limit of the constraint itself. A weak-* limit of the measures satisfies the Liouville equation of the limit flows only up to a defect, which is carried where the dissipation concentrates. We must prove that this defect is a jump term, on a measure supported on the graph of the pass map, so that the limit satisfies the Liouville equation of the limit system of Section 2.4 in the form with guard measures [5, 6], uniformly over the selections. Where nothing concentrates, the limit is of the kind that the existing theory describes, with the oscillation about the glide represented by an invariant measure [9]. The objective is met when the limit constraint is stated for the full inclusion, with each hypothesis either verified on the entry corridor or named as an assumption, together with its dual: a pair (Φ, Ψ) that is non-decreasing along the flows and across the jumps of the limit system.

1.2 The provenance of the guarantee

A polynomial barrier whose coefficients come from an interior-point solver has analytical *form*, but it inherits the solver's accuracy [12]. The documented failure modes of numerical semidefinite solvers include the loss of strict feasibility, on which the repair schemes depend [13]. A solver's output can be made into a certificate: if it is rounded to rational coefficients and re-verified exactly, it is a proof [14]. The obstacle is scale rather than principle. The rounding needs a strictly feasible Gram matrix, which a tight bound does not have, and exact rational certificates carry a bit-complexity whose exponent grows with the number of variables [15, 16]. For the thirteen-state field of Section 3 we do not pursue this route as the main line.

Instead, the certificate is to be a closed-form bound: the constants of Section 2, derived from the decomposition and evaluated over the corridor in validated interval arithmetic. No numerical optimization is the source of the guarantee.

Interval arithmetic is the setting in which a guarantee has meant, for sixty years, an enclosure rather than an approximation. It is rigorous in floating point, because every operation is rounded outward and the computed interval contains the exact result. Exact rational arithmetic would add nothing to soundness, and it cannot evaluate an exponential or a sine [17, Ch. 3]. Two distinct losses have to be managed. The first is the *dependency problem*. Interval arithmetic is inclusion isotonic, which is the property that makes an enclosure valid, but repeated occurrences of a variable are treated as independent, and distributivity weakens to subdistributivity [17, Chs. 4–5]. The second is the *wrapping effect*. The image of a box under a flow is not a box, and re-enclosing it

at every step admits points that no trajectory reaches [17, Ch. 10]. Refining the time step does not remove it. What controls wrapping is the representation of the set, for example Lohner’s QR factorization, zonotopes, or Taylor models [18, 19]. Applied interval analysis supplies the remaining machinery: certified set inversion, contractors, and validated integration [20].

Objective 2. Validated evaluation of the certificate

The guarantee is to be verified in validated arithmetic rather than produced by a numerical solver. We must show that the closed-form constants, the pass enclosures that they do not cover, and the reachable set of the limit system of Section 2.4 can be evaluated over the whole corridor with every rounding error accounted for. This is a computer-assisted step of the proof: the transcendental field and its Jacobian are enclosed with outward rounding, the corridor constraints are contracted, and the passes are integrated with validated error bounds. The limit system is itself a reachability problem, in a few slow coordinates and over every bank history. It is to be enclosed by comparison bounds in closed form where they exist, and by validated set propagation where they do not. The objective is met when every inequality that the certificate asserts is discharged by a computation that returns either a proof or an explicit failure to decide at the working precision.

2 The singular limit and its profile decomposition

This section carries the new content. The entry problem supplies a small parameter, the thin-atmosphere ratio $\varepsilon_{\text{atm}} = H_s/R_e$, where H_s is the atmospheric scale height and R_e the planetary radius. We write ε for it where the other small parameters are not in play. The question is what the family of occupation measures does as $\varepsilon \rightarrow 0$.

2.1 Loss of compactness along the time axis and the defect measure

On the corridor nothing is lost in the usual sense. The set $[0, T] \times \mathcal{X} \times \mathcal{U}$ is compact, so the occupation measures, which are nonnegative Radon measures in $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ with the weak-* topology, form a tight family and converge along subsequences. The limit is nevertheless the wrong object, because it forgets the atmospheric passes (Fig. 2.1).

How much the limit forgets depends on which limit is taken, and we address this first. A pass entered at a flight-path angle γ_e turns the velocity through an angle of about $2\gamma_e$. Since lift cannot be had without drag, it also sheds a share of the kinetic energy of order $\gamma_e/(L/D)$, where L/D is the lift-to-drag ratio. The pass crosses a scale height in a time $H_s/(V\gamma_e)$, which is of order $\varepsilon_{\text{atm}}/\gamma_e$ on the slow time scale, and it levels out where the lift is $V^2\gamma_e^2/(2H_s)$, which is of order $\gamma_e^2/\varepsilon_{\text{atm}}$ relative to gravity. The entry angle therefore enters through the ratio $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$. As in the classical Chapman–Loh variables, it is $\sqrt{\varepsilon_{\text{atm}}}$ and not ε_{atm} that sets the scale [21, 22]. Two limits must be distinguished.

- If $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ is held fixed, a pass lasts a time of order $\sqrt{\varepsilon_{\text{atm}}}$, sheds an energy of the same order, and loads the vehicle by a fixed multiple of the glide load. This is the limit of a corridor whose dynamic-pressure and load limits are held fixed. In it nothing concentrates: the passes are many and slight, and they are the troughs of an oscillation about the glide.
- If γ_e is held fixed, a pass lasts a time of order ε_{atm} and sheds a fixed share of the energy, under loads that grow like $1/\varepsilon_{\text{atm}}$. It carries occupation mass of that order, and none in the limit, while its share of the energy dissipated and of the turn of the velocity vector does not vanish. This is the limit drawn in Fig. 2.1.

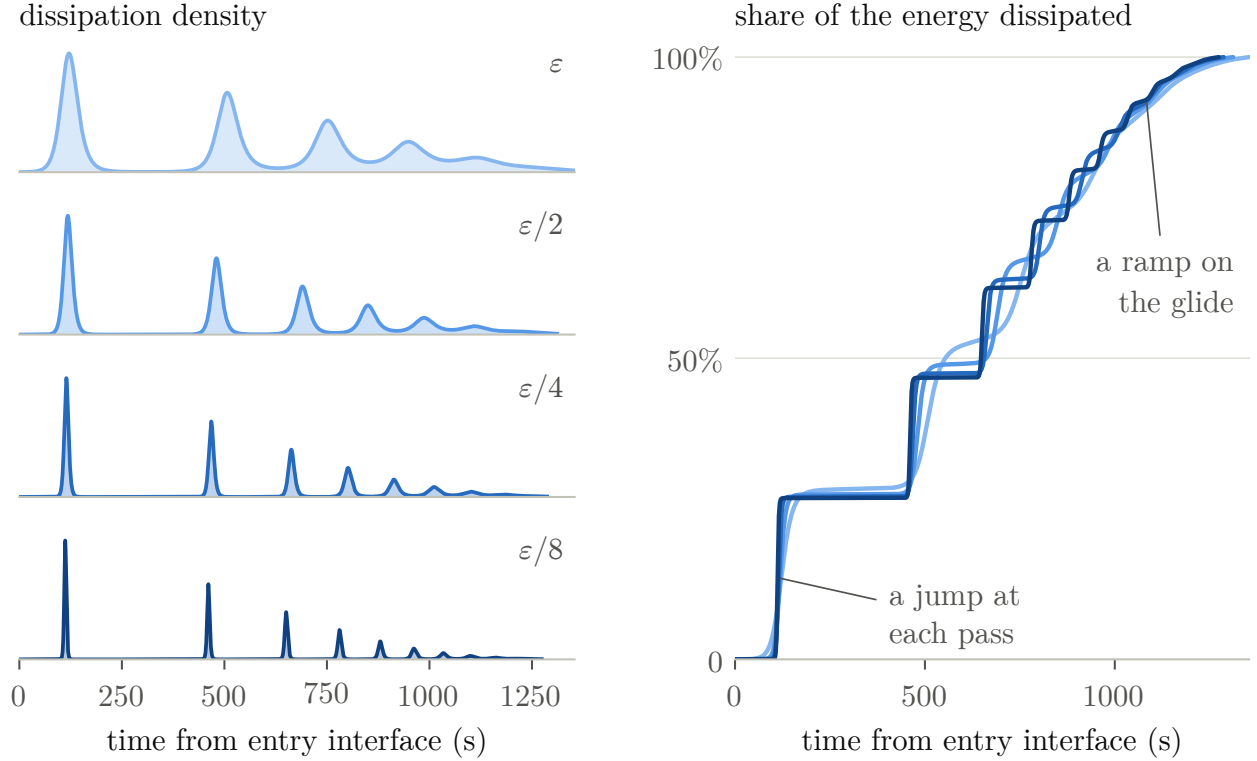


Figure 2.1: Concentration of the dissipation as the atmosphere thins, computed for illustration. A model vehicle with lift-to-drag ratio one enters at 7.6 km/s and a flight-path angle of four degrees, in an exponential atmosphere whose scale height is halved three times about a fixed altitude, so that ε is divided by two, four and eight. **Left:** the dissipation density, that is, the drag power normalized to unit mass, with each row on its own vertical scale. The passes sharpen into well-separated peaks at times that converge. **Right:** the share of the energy dissipated up to each time, from light to dark as ε decreases. The curves approach a staircase with a jump at each pass, followed by a ramp on the glide. An occupation measure weighs every interval by its length, whereas what distinguishes one trajectory from another is decided in the jumps, which have no length in the limit.

The physical case lies between the two. Since $\sqrt{\varepsilon_{\text{atm}}} \approx 0.034$ is about two degrees and entry angles are a few degrees, a pass need be neither short compared with the arc that follows it nor distinct in scale from the oscillation about the glide. The statement that we aim at is therefore one at fixed ε , with a threshold that decides which passes are resolved one by one, and the two limits are the extremes that it has to cover.

A reachable set is a support and not an integral, so what has no mass still counts. This alone does not distinguish the problem from the limit theory of singularly perturbed control, whose value functions already carry a terminal cost on the slow state [11]. The difference lies in the hypothesis under which that theory works: the fast state, and the trajectories that approximate an optimum, must stay in a bounded set, and a pass that concentrates is exactly what this excludes. We therefore read the loss where it occurs, along the fast-time axis $\tau = t/\sqrt{\varepsilon_{\text{atm}}}$. On this axis the horizon $[0, T/\sqrt{\varepsilon_{\text{atm}}}]$ exhausts the half-line and translations act non-compactly, so compactness is lost in the same way as for a problem posed on an unbounded domain.

Between passes the vehicle is outside the sensible atmosphere, on an arc governed by gravity alone. Once captured, it stays near the equilibrium-glide balance $\Lambda = 1$. In the three-degree-of-

freedom (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$, where σ is the bank angle. In the full six-degree-of-freedom formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation. The balance is then a single scalar constraint on the thirteen-dimensional translational-rotational state, and the slow set is its intersection with the trim and quasi-static-deflection conditions of the two faster motions. We take the balance with planetary rotation and oblate gravity retained. The two do not enter on equal terms: flying east at entry speed, the Coriolis term is tens of percent of the net acceleration $g - V^2/r$ that the lift has to balance, while the J_2 terms are at most a percent of it. The scalar condition $\Lambda = 1$ is the reduction under spherical gravity, no rotation, and small flight-path angle.

The loss is read on a *defect measure*, which must be chosen so that the passes carry mass. The part of the occupation-measure limit that lies off the glide cannot serve, since the passes have none. The energy dissipation can. Along every trajectory of the inclusion, the specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, changes at the rate $\dot{\mathcal{E}} = -P$, where $P = DV/m \geq 0$ is the drag power, because neither the lift nor the Coriolis force does work. Here Φ_g is the gravitational potential, ω_E the rotation rate of the planet, and ϕ the latitude. Consequently, the total dissipation satisfies $\int_0^T P dt \leq \mathcal{E}_{\max} - \mathcal{E}_{\min}$, uniformly over controls, over selections of the interval field, and over ε . We define

$$\nu_\varepsilon(\tau) = \frac{\sqrt{\varepsilon} P(\sqrt{\varepsilon} \tau)}{\int_0^T P dt}, \quad (2.1)$$

the dissipation density on the fast-time axis, normalized to unit mass and extended by zero. We propose ν_ε as the defect measure. Where it concentrates is where the limit of the occupation measures can fail to satisfy the Liouville equation of the limit flow, and the mass that it puts there should bound the size of the failure, because the aerodynamic impulse over any interval is bounded by the energy dissipated in it.

Along any sequence $\varepsilon_n \rightarrow 0$, with a trajectory chosen for each, the ν_ε are nonnegative densities of fixed mass on the line. This is exactly the hypothesis of the concentration-compactness lemma of Lions [23, Lemma I.1], which classifies such sequences into three alternatives. Here each of them has a physical meaning:

- *vanishing*: no window of fixed fast-time width holds a fixed share of the dissipation. This is the glide and the oscillation about it, on which the drag is bounded and such a window holds a share of order $\sqrt{\varepsilon}$;
- *compactness*, up to translation: one pass carries the dissipation;
- *dichotomy*: the dissipation splits between pieces whose fast-time separation diverges, that is, two passes a positive time apart, or a pass followed by a glide.

Which alternative occurs is decided by the scaling above. With the loads held fixed, the drag power is bounded and every sequence vanishes. Concentration requires an entry angle that stays large compared with $\sqrt{\varepsilon}$. In that case translations along the fast-time axis separate the passes, and the dilation half of the principle, the limit-case modification [24], resolves each pass at its own width, since a pass shorter than $\sqrt{\varepsilon}$ is a Dirac mass at that scale. The trichotomy is Lions' lemma applied as stated. That the vanishing part is the glide and the concentrated pieces are the passes remains to be proved, starting from the inner and outer expansions of the classical theory [25].

Measures that keep track of concentration are not new to control theory. An unbounded control that enters the dynamics affinely is relaxed to a measure in time whose atoms are jumps of the

state, and oscillation and concentration of a control sequence have been treated together through the measures of DiPerna and Majda [26, and the references there]. In that setting the concentration is put in by the control. Here the control is bounded and the limit produces the concentration, through a force that depends on the state, so the effect of a jump is the solution of an inner problem. Bounding the impulse alone would give a limit that is sound but far too large.

2.2 Skeleton, bubbles, residual

Rather than restore compactness, we extract the profile decomposition that iterating the dichotomy provides:

- a low-dimensional *skeleton*, which is what survives the limit: the exo-atmospheric arcs between passes, which are conics up to the oblateness correction, followed by the equilibrium glide, a graph over (r, V) in the corridor subspace once the balance is solved for the vertical lift fraction. For the glide, a rigorous characterization with quantitative accuracy estimates already exists [27]. It treats the glide as a *regular* perturbation: the exact solution is not attracted to the zeroth-order one, a point that returns in [Section 2.4](#);
- one *boundary-layer bubble* per atmospheric pass, the analogue of a concentrating profile: the pass rescaled to its own width. Its net effect on the slow coordinates is a jump. The exit states that the bank history and the selections can produce from one entry state form a compact set, and the map that assigns this set to the entry state, the *pass map* J , is the object to be enclosed. Orthogonality is read off the scale-and-core separation condition that governs concentrating profiles in general [28]. The decomposition of skip entry into an aerodynamic inner region and a gravitational outer one is classical [21, 25], and it has been carried to second order and to three dimensions [29]. These are expansions of a single trajectory, with the error assessed by example. What is not classical is a statement that is uniform over a family of trajectories, with an explicit remainder at fixed ε ;
- a *residual*, which is what the finitely many profiles do not capture. In a profile decomposition the remainder is not compact and is not small in the conserved norm; it is small in a weaker norm, and only as the number of extracted profiles grows [30, 31]. Here the weaker norm is the largest share of dissipation held by any fast-time window: on a glide the residual carries most of the dissipation, but almost none of it in any one window. The quantitative bound that we need is a different statement, a distance in state space, which we take up in [Section 2.4](#).

We keep two kinds of statement apart, because they come from different arguments. A decomposition obtained by compactness is a statement about subsequences. What it gives uniformly, it gives by contradiction: it says which pieces can occur and that nothing else can, and it produces no constant. The wave-equation theory is explicit about this, since the uniform bound that follows from its decomposition comes with a constant that its authors say they cannot describe [30, Corollary 2]. We therefore ask compactness only for the structure. The constants are to come from direct estimates: the energy identity, the bound that it gives on the aerodynamic impulse, a matching estimate at each pass, and averaging over the oscillations that do not concentrate. Each of these is an inequality with a constant that can be written down, and writing them down is [objective 6](#). The simplest of them is carried out in [Section 2.5](#).

The number of bubbles is bounded *a priori*, and the bound comes from the same monotone quantity. A pass that dissipates at least δ of specific energy can occur at most

$$K_\delta \leq \frac{\mathcal{E}_{\max} - \mathcal{E}_{\min}}{\delta} \quad (2.2)$$

times, for every control and every selection of the interval field. Passes below the threshold are left to the oscillation about the glide and to the residual. This is the device of the profile-decomposition literature: only finitely many profiles exceed any given size. The integrated heat-load budget gives the count its physical interpretation, since a trajectory can only afford so many skips [25, 32]. However, that budget depends on a stagnation-point heating correlation, whereas the energy budget does not.

Enclosing a pass is where the wrapping effect has to be met. For linear systems it can be avoided altogether: support-function methods propagate each direction independently, so successive over-approximations do not compound, and what remains is a discretization error of first order in the time step [33]. Nothing comparable is available for a nonlinear pass.

Objective 3. Wrapping-effect suppression in boundary-layer bubbles

Each boundary-layer bubble represents one atmospheric pass, and its net effect on the slow coordinates must be enclosed tightly enough that a sequence of K_δ passes does not exhaust the corridor. Wrapping in a pass has a specific source. Trajectories of a set enter the layer at different times, the set is sheared along the flow, and re-enclosing the sheared set in a box at every step admits points that no trajectory reaches. We must therefore enclose the pass map, from entry state to exit state in the rescaled time of the profile, as a single object, using a zonotope or Taylor-model enclosure of a map rather than a tube of boxes, and compose pass maps with the outer flow. The objective is met by a bound on the growth of the enclosure per pass that is explicit in the interval widths, verified on a multi-pass corridor against sampled trajectories.

2.3 Separation of scales, and what does not concentrate

Two things must be proved before the decomposition is a bound, and they differ in kind.

The first concerns the passes. In the wave-equation theory, two profiles are orthogonal when their scales diverge or, at equal scale, when their centers separate by a diverging number of widths [30, (1.18)]. Here the scale of a profile is the width of a pass and its center is the time at which the pass occurs, so passes at distinct times are orthogonal without further argument. Since the dissipation is a measure, it splits *additively* across orthogonal profiles, which is the analogue of the energy identity of that theory. The state does not split additively, because an ordinary differential equation composes: the state leaving one pass is the initial datum of the next arc. What must be proved is the analogue of the nonlinear statement of that theory, that the contributions of separate profiles do not interact up to a small error [30, (1.26)]. Here it reads as follows: the map from entry data to exit data factors through the individual pass maps and the flows between them, up to a remainder that is explicit and uniform over the family. This is a matching estimate between inner and outer solutions. The classical theory supplies its form for one trajectory [25, 29], and its small parameter is that of Section 2.1: inside a pass, gravity is negligible compared with the aerodynamic force in proportion as $\gamma_e^2/\varepsilon_{\text{atm}}$ is large, which is why a pass below the threshold cannot be treated as a jump. For the passes, the argument worth transporting from the wave-equation theory is the constructive one. Splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so that orthogonality is produced by the construction and not assumed [30].

Objective 4. Composition of passes

We must prove a non-interaction theorem for passes separated in time: a trajectory with several passes factors through the pass maps and the flows between them, for every selection of the interval field, with a matching error at each pass whose constant is in closed form. The statement has a range of validity, namely $\gamma_e^2/\varepsilon_{\text{atm}}$ above a threshold, and through the relation between the turn of a pass and the energy that it sheds, that threshold is the δ of Eq. (2.2). Passes below it are not composed but averaged. The objective is met by the factorization with its remainder, for the translational state under interval disturbances, verified against sampled multi-pass trajectories.

The second concerns what does not concentrate. The attitude and the elastic motions are not events. They are oscillations, at the short-period attitude and first structural periods T_{att} and T_{ela} , measured against the slow time $T_{\text{slow}} = \sqrt{R_e/g}$ by $\varepsilon_{\text{att}} = T_{\text{att}}/T_{\text{slow}}$ and $\varepsilon_{\text{ela}} = T_{\text{ela}}/T_{\text{slow}}$, and they persist for as long as there is air. The same holds for the oscillation about the glide whenever the loads are bounded. In a profile decomposition, motions of this kind are not profiles. They remain in the remainder, which the wave-equation theory is careful to say is not small in energy, since it contains every wave that concentrating profiles cannot represent [30, p. 136]. The tool for them is averaging [34, 35]: each oscillation is carried by a slowly varying amplitude, and the slower dynamics see it through its mean.

Two difficulties stand between this tool and a bound. First, an averaging theorem gives the error to an order, and an order is not a bound, as the standard text says of its own estimates [34, Sec. 10.1]. Second, with several oscillations the theorem holds only away from resonance, and the frequencies here are not constants. The attitude frequency grows with the square root of the dynamic pressure, so it rises from zero at the start of every pass, and all the frequencies are known only to within the aerodynamic intervals. Where two frequencies cross, averaging over both phases fails in a zone about the resonance. The treatment there is itself an inner expansion matched to an outer one, and a complete theory is not available even for a single smooth field [35, Chs. 7–8]. The results that pass through a resonance without resolving it hold, where capture is possible, outside a set of initial conditions of small measure, and an outer bound can discard nothing. This is the one genuinely new obligation for which no theory is at hand, and it sits on the critical path.

Objective 5. Separation of scales under interval disturbances

The framework separates three scales: atmospheric, attitude and elastic. The oscillations that do not concentrate, namely the oscillation about the glide, that of the attitude and that of the structure, must be averaged with an error that is explicit, uniform over the selections of the interval field, and valid for every trajectory, including those that pass through a resonance. We must prove an averaging theorem with constants. The amplitude of each oscillation is carried as a slow coordinate of the limit system of Section 2.4 or enclosed in the interval field, and the theorem bounds the distance between the slow coordinates of the full system and those of the averaged inclusion in closed form. The interval widths enter twice: as a disturbance of each oscillation, and through the frequencies, which depend on the uncertain coefficients, so that the ordering of the frequency bands is itself a hypothesis to be verified over the corridor. Where two bands cross, the statement is for the coupled pair, carried jointly through the crossing, with the width of the resonance zone and the change of amplitude across it enclosed. The objective is met by the theorem for the oscillation about the glide and for the attitude, with the remainder in closed form, and with the elastic modes either included or enclosed as a widened interval, as in the fallback of Section 4.

2.4 The certified bound

The skeleton and the bubbles together are not a surface with corrections but a system of its own, with flows and jumps, and we state the target against it. In the slow coordinates, the *limit system* $\mathcal{H}_\delta$ has two flows and one jump. On an arc the state moves under gravity alone. On the glide it moves along the balance $\Lambda = 1$, under the admissible bank histories and the selections of the interval field, and it carries one additional coordinate, the amplitude of the oscillation about the balance. A pass that dissipates more than the threshold δ of Eq. (2.2) is a jump: the state that enters it is sent to the set $J(x)$ of states that can leave it, onto an arc or onto the glide, with the time and the range consumed by the pass included, and a trajectory makes at most K_δ jumps. The target statement is

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\Pi(\mathcal{R}_{\mathcal{H}_\delta}(T))}_{\text{skeleton and bubbles}} \oplus \underbrace{\mathbb{B}(\varrho(\varepsilon, \delta))}_{\text{residual}}, \quad (2.3)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. Here $\mathcal{R}_\varepsilon(T)$ is the set of slow states reachable at time T from the initial set, under every admissible control and every selection of the interval field, and $\mathcal{R}_{\mathcal{H}_\delta}(T)$ is the corresponding set for the limit system. The map Π reads a state of the limit system back in the coordinates of the full one: it is the identity on an arc, and on the glide it returns the altitudes and flight-path angles that an oscillation of the carried amplitude visits. The symbol $\oplus$ denotes the Minkowski sum, and $\mathbb{B}(\varrho)$ is the ball of radius ϱ in the corridor metric, a weighted supremum norm in which altitude is measured in scale heights and each remaining slow coordinate in units of its corridor width. Passes compose in sequence, as jumps do, and their effects are not added, since an arc that meets the atmosphere has no continuation without the pass. Figure 2.2 illustrates the statement.

Equation (2.3) is a statement about sets. As a statement about measures, it is the claim of **objective 1**: the occupation measures converge to measures that satisfy the Liouville equation of $\mathcal{H}_\delta$, with the mass that crosses each pass carried by a measure on the graph of J . The dual pair (Φ, Ψ) of Section 1.1, non-decreasing along the two flows and across the jump, is the form in which an outer bound on $\mathcal{R}_{\mathcal{H}_\delta}(T)$ is certified.

The glide must carry an amplitude because of the dynamics. Let z denote the altitude above the glide in scale heights, and let $G = g - V^2/r$ be the acceleration that the lift has to balance. Lift falls off exponentially with altitude and G does not, so to leading order in ε , at frozen speed and with the damping neglected,

$$\ddot{z} = \frac{G}{H_s} (e^{-z} - 1). \quad (2.4)$$

The glide is the equilibrium of this equation. The phugoid, the oscillation of altitude against flight-path angle about the glide, is its small oscillation, with frequency $\sqrt{G/H_s}$, which grows like $\varepsilon^{-1/2}$. A skip is its large oscillation: a rebound from the exponential wall below and a ballistic arc under $-G$ above, which is how the classical theory describes skip paths, as oscillatory paths of large amplitude [25]. An oscillation that crosses the glide altitude at a flight-path angle γ_0 has an amplitude fixed by γ_0^2/ε , the parameter of the scaling in Section 2.1. The two limits distinguished there are thus the two ends of one family, and the threshold δ is a cut in the amplitude: an oscillation above it is resolved pass by pass, and one below it is averaged.

The terms neglected in Eq. (2.4) decide the analysis. The damping rate of the phugoid is $g/[V(L/D)\cos\sigma]$. It does not depend on ε , while the frequency grows like $\varepsilon^{-1/2}$. In the limit, the motion normal to the glide is therefore an undamped oscillation on its own time scale. The glide balance is not a normally hyperbolic slow manifold in the sense required by the geometric theory [37], and no boundary-layer estimate of Tikhonov type brings a trajectory onto it. Moreover, the amplitude is not small. A change of bank moves the glide altitude by a fixed fraction of a scale

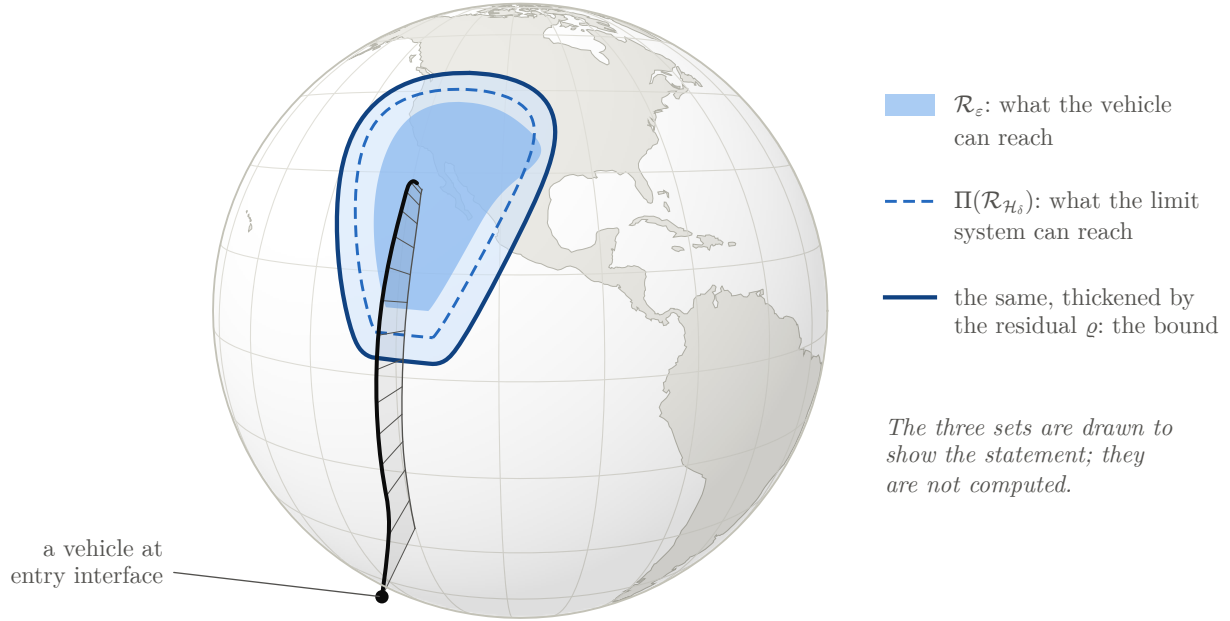


Figure 2.2: Illustration of the target statement Eq. (2.3), drawn on the ground. The trajectory is an example, a lunar-return skip entry simulated from a published entry state [36], with altitude drawn twelve times too large and a post marking each minute. The three sets are an illustration and are not computed: the set that the vehicle can reach, the set that the limit system can reach, and the latter thickened by the residual, which is the bound. The first may leave the second, but by no more than the residual.

height, however thin the layer, and a bank history in step with the oscillation pumps it, just as one opposed to it damps it. Guidance laws use the latter [38], and a reachable set must allow for the former. The amplitude is therefore a coordinate on which the control acts, and its range belongs to $\mathcal{R}_{\mathcal{H}_\delta}$, not to the residual.

The residual $\varrho(\varepsilon, \delta)$ collects what the limit system does not carry. It has three parts: the lag of a trajectory behind the descending glide, the matching error at each of at most K_δ passes, and the averaging error of the oscillations. The residual and the count K_δ are to be derived as explicit, closed-form expressions in the corridor bounds and the aerodynamic interval widths, independent of any numerical optimization. One piece is immediate: the count is given by Eq. (2.2). The rest must be derived. For the lag, we expect a bound of order $\sqrt{\varepsilon}$ in scale heights, the order of the damping ratio of the phugoid, with a constant depending on the speed, the lift-to-drag ratio and the bank angle that degenerates as V approaches circular speed, where the glide gives way to arcs and passes. Such a bound describes the ringing amplitude of a trajectory started level on the glide and nothing more. It says nothing about a trajectory that leaves a pass, which rings with whatever amplitude the pass left it, for a time comparable to the entry. That amplitude is what the limit system carries.

In the spherical-LVLH coordinates the altitude and speed bounds of the corridor are box constraints on (r, V) . The heating, dynamic-pressure and load limits are not: each is a curve $h \geq h_{\min}(V)$. Each is, however, monotone in altitude and in speed at fixed attitude, so contracting a box against it returns the exact interval hull of the feasible part, and the interval widths that enter ϱ and K_δ inherit that tightness. Requiring the constants in closed form is not what makes the bound a proof, since a verified solver output is one too (Section 1.2). It is what makes the dependence on the corridor and on the interval widths explicit, so that the bound is evaluated again,

and not solved for again, when either changes.

Objective 6. Analytical derivation of the corridor constants

The guarantee rests on the count K_δ and the residual $\varrho(\varepsilon, \delta)$. We must derive both as explicit functions of the corridor bounds, the initial set and the aerodynamic interval widths, without relying on any numerical solver. The count follows from the energy budget, with the threshold δ chosen against the residual that it leaves behind: lowering it resolves more passes one by one and pays a matching error for each, while raising it leaves larger oscillations to be averaged. The residual has three parts: the lag behind the glide, the matching error of [objective 4](#), and the averaging error of [objective 5](#). Each constant must hold for every selection of the interval field, and its range of validity, bounded away from circular speed, is part of the statement. The objective is met by formulas whose evaluation over the corridor in validated interval arithmetic bounds the distance from sampled trajectories of the full model to the reachable set of the limit system, with no fitted number. The comparison is to be made on a benchmark, a capsule skip entry reconstructed from published data, with the tightness of the bound stated against the set-propagation baseline of [Section 3](#).

2.5 A first bound from the energy identity

One of the direct estimates can be carried out at once, and we record it for two reasons. First, it is a bound of the kind that the program asks for: a closed formula in the corridor data that holds for every admissible control and every selection of the interval field. Second, the way in which it fails shows what the decomposition has to supply.

We write $\Phi = \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$ for the potential in the rotating frame, so that the energy of [Section 2.1](#) is $\mathcal{E} = \frac{1}{2}V^2 + \Phi$, and we write $s(t)$ for the length of the ground track up to time t . Let $\mathcal{C}$ be a part of the corridor that lies below a ceiling, on which the potential is at most $\Phi_{\max}$ and the density satisfies

$$\rho \geq \rho_c \exp\left(\frac{\Phi_{\max} - \Phi}{gH}\right).$$

That is, the air at the ceiling is at least as dense as ρ_c , and below the ceiling it thickens at least as fast as in an exponential atmosphere of scale height H , where g is an upper bound for the acceleration of gravity. Let β be an upper bound for the ballistic coefficient $m/(C_D S)$, where C_D is the drag coefficient and S the reference area, so that the drag satisfies $D/m \geq \rho V^2/(2\beta)$ for every selection of the interval field.

Proposition 2.1 (Range bound). *Suppose that a trajectory of the inclusion stays in $\mathcal{C}$ on $[0, T]$, with speed V_0 at time zero and speed at least $V_f > 0$ throughout. Then the length of its ground track satisfies*

$$s(T) \leq \frac{2\beta}{\rho_c} \left[\ln \frac{V_0}{V_f} + \frac{gH}{V_f^2} \right]. \quad (2.5)$$

Proof. The ground track advances no faster than the vehicle, so $\dot{s} \leq V$. Since $\dot{\mathcal{E}} = -(D/m)V$ and the drag is positive, we have $\dot{s} \leq -\dot{\mathcal{E}}/(D/m)$. We now bound the drag from below in terms of the energy alone. On $\mathcal{C}$ we have $V^2 = 2(\mathcal{E} - \Phi)$, so

$$\frac{D}{m} \geq \frac{\rho_c}{\beta} \exp\left(\frac{\Phi_{\max} - \Phi}{gH}\right) (\mathcal{E} - \Phi).$$

At a fixed energy the right-hand side decreases as Φ increases, since both of its factors do. The potential is at most $\Phi_{\max}$, and it is at most $\mathcal{E} - \frac{1}{2}V_f^2$ because $V \geq V_f$. We set $\mathcal{E}^* = \Phi_{\max} + \frac{1}{2}V_f^2$. For $\mathcal{E} \geq \mathcal{E}^*$ the right-hand side is at least its value at $\Phi = \Phi_{\max}$, and for $\mathcal{E} \leq \mathcal{E}^*$ it is at least its value at $\Phi = \mathcal{E} - \frac{1}{2}V_f^2$. Consequently,

$$\frac{D}{m} \geq \begin{cases} \frac{\rho_c}{\beta} (\mathcal{E} - \Phi_{\max}), & \mathcal{E} \geq \mathcal{E}^*, \\ \frac{\rho_c}{2\beta} V_f^2 \exp\left(\frac{\mathcal{E}^* - \mathcal{E}}{gH}\right), & \mathcal{E} \leq \mathcal{E}^*. \end{cases}$$

Since $\mathcal{E}$ decreases along the trajectory, we may take it as the variable of integration, so that $s(T)$ is at most the integral of the reciprocal of this bound from $\mathcal{E}(T)$ to $\mathcal{E}(0)$. The integrand is positive and $\mathcal{E}(0) \leq \Phi_{\max} + \frac{1}{2}V_0^2$, so the integral only increases if it is taken over all energies below $\Phi_{\max} + \frac{1}{2}V_0^2$. Above $\mathcal{E}^*$ it is

$$\frac{2\beta}{\rho_c} \int_{\mathcal{E}^*}^{\Phi_{\max} + \frac{1}{2}V_0^2} \frac{d\mathcal{E}}{2(\mathcal{E} - \Phi_{\max})} = \frac{2\beta}{\rho_c} \ln \frac{V_0}{V_f},$$

and below $\mathcal{E}^*$ it is

$$\frac{2\beta}{\rho_c V_f^2} \int_{-\infty}^{\mathcal{E}^*} \exp\left(-\frac{\mathcal{E}^* - \mathcal{E}}{gH}\right) d\mathcal{E} = \frac{2\beta}{\rho_c} \frac{gH}{V_f^2}.$$

Adding the two contributions gives [Eq. \(2.5\)](#). □

The two terms have a simple meaning. The first is the distance in which drag alone slows the vehicle from V_0 to V_f in level flight at the ceiling. The second accounts for the potential energy, which the vehicle can spend against the drag at its lowest speed; only the first scale height below the ceiling contributes appreciably, because the air thickens underneath. No property of the control enters, because the lift does no work, so the bound holds for every bank history. Similarly, the interval field enters only through ρ_c and β .

We evaluate [Eq. \(2.5\)](#) on the example of [Fig. 2.2](#). In that simulation the vehicle descends through 40 km for the last time at $V_0 = 2.34$ km/s, and we take $\mathcal{C}$ to be the states below that altitude, down to $V_f = 305$ m/s. The model vehicle has $\beta = 281$ kg/m². For the air we take the standard atmosphere [\[39\]](#), whose density below 40 km increases downward at least as fast as an exponential of scale height $H = 7.0$ km, with $g = 9.9$ m/s². For ρ_c we take its density at 40 km reduced by fifteen percent, that is, $\rho_c = 3.4 \times 10^{-3}$ kg/m³. The reduction covers the variation of the potential with latitude along the ceiling, and it leaves room for an atmosphere up to twelve percent thinner than the standard one. Then $2\beta/\rho_c = 165$ km and $gH/V_f^2 = 0.75$, so that

$$s(T) \leq 165 \text{ km} \times [2.04 + 0.75] \approx 460 \text{ km}.$$

Every bank history that keeps the vehicle below 40 km therefore lands within this distance of the point over which it started ([Fig. 2.3](#)). For comparison, a sweep of bank histories simulated from the same state, each of which stays below that altitude, lands within 113 km.

The bound is certified and uniform over the family, and it is too large by a factor of about four. The reason is instructive. Nearly three quarters of [Eq. \(2.5\)](#) is its first term, which lets the vehicle slow down at the ceiling, in the thinnest air of $\mathcal{C}$. No vehicle can hold that altitude as it slows, since its lift falls with the square of the speed, but the energy identity does not know this, because the lift does no work. The lift constrains the range only through the balance normal to the path, that is, through the glide. More importantly, the bound is proportional to $1/\rho_c$, so it degenerates as the ceiling of $\mathcal{C}$ is raised. Evaluated from the states at which the same flight descends through 45 km and through 50 km, it is 1,070 km and 2,280 km: it more than doubles with every five kilometres. It

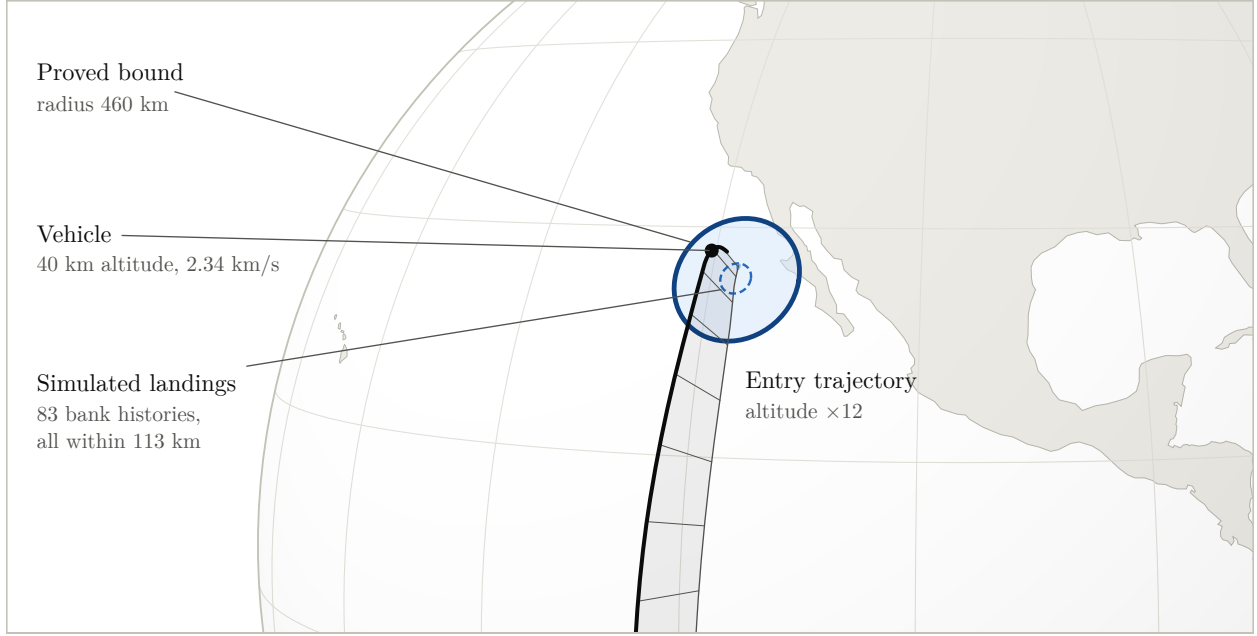


Figure 2.3: The bound of [Proposition 2.1](#), evaluated on the example. The view is that of [Fig. 2.2](#), enlarged about the end of the entry, so that the entry interface is out of the frame; the trajectory is the same simulated skip entry, with altitude drawn twelve times too large and a post marking each minute. From the state marked *Vehicle*, at which the vehicle last descends through 40 km, every bank history that keeps it below that altitude lands inside the solid circle. The landing points of a sweep of simulated bank histories from the same state lie inside the dashed circle. The solid circle is proved, and the dashed one is a sample.

says nothing about a trajectory that leaves the atmosphere between two passes, for which there is no positive ρ_c . This is the difficulty of the thin-atmosphere limit in its simplest form: an estimate that treats the trajectory as a whole cannot be uniform. The remedy is to estimate each piece on its own scale, the arcs by gravity alone, each pass through its own map, and the glide through its balance, which is the decomposition of [Section 2](#).

3 The application: hypersonic entry

Entry is not an arbitrary illustration of the machinery above. It supplies the singular perturbation, the limit system and the concentrating events in one system, and, for the reasons that follow, the existing methods stop short of it. The vehicle and its flight corridor enter the framework only as data, so the bound is sought for hypersonic entry vehicles in general and not for a particular one.

3.1 Why existing methods do not reach it

The direct method solves a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in the state dimension and capped in practice near four or five states [\[40, 41\]](#). A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (3.1)$$

and more once the heat load, the mass, the thermal response and recession of the heat shield and the aerothermoelastic modes are coupled [42]. The two counts use “degree of freedom” in its two customary senses. The vehicle has six degrees of freedom in the configuration sense (three translational, three rotational), while Eq. (3.1) has thirteen coordinates spanning a twelve-dimensional state manifold once the unit-quaternion constraint $\|\vec{q}\| = 1$ is imposed. So the exact grid method quits before the real problem starts.

Set-propagation methods go further. Taylor-model and zonotope flowpipes return rigorous outer bounds at ten to twenty states, and they are the baseline against which this program must be measured [19, 43, 44]. What they return is an enclosure of one instance: a numerical set for one vehicle, one initial box and one horizon, whose tightness tends to degrade with the width of the initial set and with the length of weakly damped motion that it has to follow, and the phugoid is weakly damped for the whole of the glide. They are the right tool inside a pass, where the horizon is short, and this is how [objective 3](#) uses them. They do not produce a bound whose dependence on the corridor and on the interval widths is explicit, which is what a statement uniform over a family requires.

In practice the question is answered by sampling. Entry guidance has converged on numerical predictor–corrector algorithms whose performance is established by simulation over dispersed cases [38], which is a sample of the family and not an enclosure of it.

What makes the measure-theoretic route viable is not that it evades the dimension but that the bound it produces does not scale with it: the limit system is low-dimensional by construction, each pass enters through one map, and the residual is a single ball.

3.2 The state space and the corridor

We carry the translational state in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) . In these coordinates the altitude and speed bounds of the entry corridor are box constraints on states, and the heating, dynamic-pressure and load limits are monotone in the two coordinates that they bind, which is the natural setting for interval contraction. Attitude is a unit quaternion, free of Euler-angle singularities. The chart has a price. The longitude and heading rates are singular at the geographic poles, so we take the corridor with $|\phi| \leq \phi_{\max} < \pi/2$, and a trajectory class that overflies a pole needs a second chart or a Cartesian formulation.

Because interval arithmetic evaluates transcendentals natively, the algebraic adjunctions that recast the system in polynomial form [45] are not needed. The state space therefore carries only the physical dimensions (spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states), with no artificial embeddings. In particular, the two auxiliary states of the polynomial formulation, the density root $y = \sqrt{\rho/\rho_0}$ and the wall-temperature surrogate, are eliminated. This is a design decision rather than an open problem, and its trade-off is the following: transcendental dynamics in coordinates native to the corridor and the glide, against polynomial dynamics in which the corridor is a nonlinear function of the state.

4 Plan of work

Six concrete obligations carry the program. They are stated where each arises above, and we gather them here, in order, as the plan of work.

1. *Liouville equation for the interval inclusion*: what the occupation measures of the interval inclusion converge to, and the Liouville constraint, with its jump term, that the limit satisfies.

2. *Validated evaluation of the certificate*: interval evaluation, contraction and integration that discharge the certificate with every rounding error accounted for, and that enclose the reachable set of the limit system.
3. *Wrapping-effect suppression in boundary-layer bubbles*: tight enclosures carried through a sequence of atmospheric passes without premature blow-up.
4. *Composition of passes*: the factorization of a trajectory through the pass maps and the flows between them, with an explicit matching remainder.
5. *Separation of scales under interval disturbances*: an averaging theorem with constants, uniform over the selections and valid where frequencies cross. This is **the critical path**, on which the reduction to a limit system and the closed-form bound rest.
6. *Analytical derivation of the corridor constants*: the count K_δ and the residual $\varrho(\varepsilon, \delta)$ from the physical corridor and the interval widths, free of any numerical solver.

Objective 1 states the limit; objectives 3 to 5 establish the decomposition; objective 6 reduces it to closed form; and objective 2 evaluates it. Objective 5 gates the rest. The central theorem is open, and apart from the elementary bound of Section 2.5 no part of this program is claimed as done.

4.1 Sequencing

The dependency structure suggests three phases rather than six parallel tracks. The first phase establishes the embedding, that is, objective 1 together with the certified envelope of the atmospheric density that it needs in order to be stated on a concrete inclusion. The second phase attacks the critical path, objective 5. This is where the program either works or does not, so it should be reached early enough that a negative result is still recoverable. Its first case is small and self-contained: the planar oscillator of Eq. (2.4), with its damping, its drift, and a bank history acting on it. Objectives 3 and 4 run alongside, since the matching argument and the enclosure geometry constrain each other. The third phase reduces the decomposition to closed form (objective 6) and completes the validated evaluation of objective 2. The validated evaluation is the one part of the program that a weaker theorem does not shorten.

We identify two retreat positions now. If objective 5 resists a proof where frequencies cross, the fallback is a corridor on which the frequency bands are ordered (a floor on the dynamic pressure and a ceiling on the speed below circular), with the crossings at the edges of each pass left to the pass enclosures of objective 3, which integrate the coupled field and need no separation, and with the elastic modes treated as a widened interval. This weakens the residual without invalidating the decomposition. If the amplitude of the oscillation about the glide cannot be bounded over every bank history, we state the result for bank histories that vary slowly compared with the oscillation, for which the amplitude decays. This restricts the family and keeps the bound explicit.

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